

Agenda

1. Announcements

- Module Quiz 16 due tonight
- 9.5 and 11.1 HW due Wednesday
- Discussion Quiz 6 on Thursday over
- Mid-semester feedback: go.wisc.edu/udahj3
- I won't have office hours this week

2. Section 11.2

Recall:

11.1 Sequences: $a_1, a_2, a_3, \dots, a_n, \dots = \{a_n\}_{n=1}^{\infty}$

11.2 Series: $a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n$

partial sums: add the first n :

$$a_1 + a_2 + a_3 + \dots + a_n = S_n = \sum_{i=1}^n a_i$$

Sequence of partial sums: $S_1, S_2, S_3, \dots, S_n, \dots = \{S_n\}_{n=1}^{\infty}$

so,

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i = \boxed{\lim_{n \rightarrow \infty} S_n}$$

We can identify whether this converges or diverges.

Big Picture

Important Series

Geometric Series Let a, r be constants where $a \neq 0$.

We call a series in the form $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \dots$ a geometric series.

We can find a formula for the n^{th} partial sum of a geometric series:

$$S_n = a + ar + ar^2 + \dots + ar^{n-2} + ar^{n-1}$$
$$rS_n = ar + ar^2 + ar^3 + \dots + ar^{n-1} + ar^n$$

$$S_n - rS_n = a - ar^n$$

$$S_n(1-r) = a(1-r^n)$$

$$S_n = a \left(\frac{1-r^n}{1-r} \right)$$

We can use $\lim_{n \rightarrow \infty} S_n$ to determine when $\sum_{n=1}^{\infty} ar^{n-1}$

Converges or diverges.

1.) If $-1 < r < 1$, then $\lim_{n \rightarrow \infty} r^n = 0$, so $\lim_{n \rightarrow \infty} a \left(\frac{1-r^n}{1-r} \right) = a \left(\frac{1}{1-r} \right) = \frac{a}{1-r}$

If $-1 < r < 1$, then $\sum_{n=1}^{\infty} ar^{n-1}$ converges to $\frac{a}{1-r}$
 $|r| < 1$; converges

2.) If $r \geq 1$ or $r \leq -1$, $\lim_{n \rightarrow \infty} r^n$ diverges, so $\lim_{n \rightarrow \infty} a \left(\frac{1-r^n}{1-r} \right)$ diverges

If $r \geq 1$ or $r \leq -1$, then $\sum_{n=1}^{\infty} ar^{n-1}$ diverges
 $|r| \geq 1$; diverges

Numbers with powers of $n \rightarrow$ geo. seq.

Example: Determine whether the series converges or diverges.

1.) $\sum_{n=1}^{\infty} 3^{-n+2} 2^{n+1} = \sum_{n=1}^{\infty} ar^{n-1}$ Goal: Find a & r

Brute Force Method

$$= 3 \cdot 2^2 + 0 \cdot 2^3 + 3^{-1} \cdot 2^4 + 3^{-2} \cdot 2^5$$

$n=1 \quad n=2 \quad n=3 \quad n=4$

$$= \boxed{12 + 8} + \frac{16}{3} + \frac{32}{9} + \dots$$

$$\boxed{a=12}, ar=8$$

$$\boxed{12r=8}$$

$$\boxed{r=2/3}$$

$|r| < 1 \Rightarrow 2/3$
converges

$$= 12 + 12(2/3) + 12(2/3)^2 + \dots = \sum_{n=1}^{\infty} 12(2/3)^{n-1}$$

$$\rightarrow \frac{a}{1-r} = \frac{12}{1-(2/3)} = \boxed{36}$$

2.) $\sum_{n=0}^{\infty} \frac{4^{2n}}{(-5)^{n+1}} = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$

Exponent Rules

$$= \sum_{n=0}^{\infty} \frac{(4^2)^n \cdot -5}{(-5)^{n+1} \cdot -5} = \sum_{n=0}^{\infty} (-5) \cdot \frac{16^n}{(-5)^n} = \sum_{n=0}^{\infty} (-5) \left(\frac{16}{-5}\right)^n = \sum_{n=1}^{\infty} (-5) \left(\frac{16}{-5}\right)^{n-1}$$

need 1 more
so multiply in

Geometric Series, $r = \frac{-16}{-5}$, $|r| \geq 1$, diverges

make clear all of this

3.) $\sum_{n=3}^{\infty} 7 \left(\frac{1}{3}\right)^{n-1} = \sum_{n=1}^{\infty} \frac{7}{9} \left(\frac{1}{3}\right)^{n-1}$

$$= \frac{7(\frac{1}{3})^{3-1}}{a} + \frac{7(\frac{1}{3})^{4-1}}{ar} + \dots = \frac{a}{1-r} = \frac{7/9}{1-1/3}$$

$$a = 7/9 \quad ar = 7/9 \cdot \frac{1}{3} \quad r = 1/3$$

Geometric Series, $r = 1/3$, $|r| < 1$,
converges to $\frac{7/9}{1-1/3}$

Harmonic Series: $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \dots$$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$

$$> 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots = 1 + \sum_{n=1}^{\infty} \frac{1}{2} \text{ diverges}$$

Therefore the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

no powers of n , partial frac. decomp

Telescoping Series

Example: $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \dots$

$$\frac{1}{n(n+1)} = \frac{A}{n} + \frac{B}{n+1} \quad (\dots) \quad \begin{matrix} A=1 \\ B=-1 \end{matrix}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$S_n = \left(\cancel{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{n-1} - \frac{1}{n} \right) + \left(\frac{1}{n} - \cancel{\frac{1}{n+1}} \right)$$

the only things that don't cancel.

$$S_n = 1 - \frac{1}{n+1}$$

$$= \lim_{n \rightarrow \infty} S_n = \boxed{1} \therefore \text{converges}$$

could be more than
just 1 on each end,
but middle always
cancels

Properties of Series

Let c be a constant. Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge.

$$1.) \sum_{n=1}^{\infty} c a_n = c \sum_{n=1}^{\infty} a_n$$

$$2.) \sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

only works if both converge

Theorem: If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Contrapositive is verf.

Divergence Theorem: If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges

Activity: Determine whether the series converges or diverges.

1.) $\sum_{n=1}^{\infty} \frac{n}{2n+1}$

2.) $\sum_{n=1}^{\infty} \sin n$

3.) $\sum_{n=1}^{\infty} \frac{1}{n}$